Transactions of the Royal Society of Tropical Medicine and Hygiene (2008) **102**, 591—598

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Characterisation of DDT, pyrethroid and carbamate resistance in *Anopheles funestus* from Obuasi, Ghana

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Received 26 October 2007; received in revised form 26 February 2008; accepted 26 February 2008
Available online 10 April 2008

KEYWORDS
Anopheles funestus;
Sodium channel gene;
DDT;
Pyrethroids;
Carbamates;
Ghana

Summary Indoor-resting anopheline mosquitoes were collected from Obuasi, Ghana, and were identified morphologically and by PCR as *Anopheles funestus* Giles. Wild-caught females were induced to lay eggs. Samples of F1 progeny from each family were divided into cohorts and were either exposed to DDT and permethrin or were stored for biochemical analysis. Bioassay data by family show evidence of DDT and pyrethroid resistance in the parent *A. funestus* population. The sodium channel gene of DDT survivors and DDT-susceptible individuals was PCR amplified and sequenced to determine whether any *kdr*-type mutations were present. Molecular analysis of the IIS5—IIS6 segment of the sodium channel gene gave no indication of any *kdr*-type mutations associated with resistance phenotypes. Biochemical analysis suggests that DDT and pyrethroid resistance may be metabolically mediated, although there were no clear correlations between enzyme levels/activities and insecticide resistance across families. Furthermore, an altered acetylcholinesterase conferring carbamate resistance was evident. These results can be used to plan an effective malaria control strategy in the Obuasi region.
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1. Introduction

Anopheles funestus Giles is an efficient African malaria vector with a malaria transmission potential that may exceed that of *A. gambiae* Giles in some cases (Fontenille et al., 1997; Gillies and De Meillon, 1968). Although distantly related, both species are especially susceptible to control by indoor application of insecticides owing to their pronounced anthropophilic and endophilic tendencies. Insecticide application by indoor residual spraying has proven efficacy for malaria vector control based on the assumptions that target vector populations retain their susceptibility to insecticide exposure and that insecticides are used reliably,

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0035-9203/$ — see front matter © 2008 Royal Society of Tropical Medicine and Hygiene. Published by Elsevier Ltd. All rights reserved.
doi:10.1016/j.trstmh.2008.02.022

efficiently and sufficiently in terms of coverage. The use of insecticide-treated fabrics (bed nets, curtains, sheeting, hammocks) may also produce significant levels of protection against malaria infection, the assumptions being that target mosquito populations retain susceptibility to pyrethroids, that coverage is adequate and that fabrics are used appropriately.

Insecticide resistance in *A. gambiae* is widespread in the West and East African regions. Designing manageable solutions to combat resistance as well as to limit the spread of resistance in this species is likely to be compounded by the emergence of insecticide resistance in sympatric *A. funestus* populations. Reports of insecticide resistance in *A. funestus* to date include resistance to dieldrin in Benin, Cameroon, Ghana, Kenya, Mali and Nigeria (Brown, 1986), to DDT and bendiocarb in Ghana (Coetzee et al., 2006) and to pyrethroids and some carbamates in South Africa and southern Mozambique (Brooke et al., 2001; Casimiro et al., 2006; Hargreaves et al., 2000).

The mechanisms by which mosquitoes become resistant to insecticides are primarily based on increased metabolic detoxification and/or decreased target site sensitivity (also called knockdown resistance (*kdr*) in the case of a particular DDT/pyrethroid resistance phenotype) (Hemingway and Ranson, 2000; Liu et al., 2006). In most cases, one or multiple metabolic pathways (mediated by glutathione S-transferases (GST), monooxygenases and/or esterases) produce resistance phenotypes that may be enhanced by *kdr*-type mutations. On the other hand, there is some doubt as to whether *kdr* mutations alone can produce a resistant phenotype. Knockdown resistance is generally described as the effect of a single amino acid substitution in domain II of the sodium channel gene and has been closely associated with DDT and pyrethroid resistance in *A. gambiae* (Martinez-Torres et al., 1998; Ranson et al., 2000). However, Matambo et al. (2007) demonstrated the presence of the Leu→Phe *kdr* substitution in a laboratory strain of the malaria vector *A. arabiensis* from Sudan and showed that the mutation assorts independently of any insecticide resistance phenotype. To date, there have been no reports of *kdr* mutations in *A. funestus*.

We report on the characterisation of DDT, pyrethroid and carbamate resistance in a population of *A. funestus* from the Obuasi region of Ghana. Indications of DDT and carbamate resistance in this population were originally reported by Coetzee et al. (2006), although their data gave no indication of pyrethroid resistance.

2. Materials and methods

2.1. Mosquito field collections

Mosquito samples were collected resting indoors in the Obuasi region of Ghana (06°15′ N, 01°36′ E) during February 2005. Wild-caught females morphologically identified as members of the *A. funestus* group (Gillies and Coetzee, 1987; Gillies and De Meillon, 1968) were later identified to species level using the PCR assay of Koekemoer et al. (2002).

Wild-caught females were transported to the Vector Control Reference Unit (VCRU) of the National Institute for Communicable Diseases (NICD) in Johannesburg, South Africa. Females were induced to lay eggs and then larvae from each family were reared through to adults. F1 male and female progeny from each family were separated on emergence. Progeny by family were then divided into two cohorts per gender. One cohort was used for standard WHO insecticide susceptibility bioassays whilst the other cohort was stored at −70 °C in Eppendorf tubes for subsequent biochemical analysis of those enzyme families usually associated with insecticide resistance.

2.2. Laboratory strains of *Anopheles funestus*

FANG is a strain originating from southern Angola and kept in colony since January 2003. It is fully susceptible to all insecticides. This colony was used for comparative purposes in the biochemical and molecular analyses.

2.3. Insecticide bioassays

Cohorts of 2–3-day-old F1 males and females from each *A. funestus* family were exposed to 4% DDT and/or 0.75% permethrin for 1 h according to the standard procedure for testing adult anopheline susceptibility to insecticides (WHO, 1998). Knockdown was recorded after 1 h and final mortality was recorded 24 h post exposure.

To assess the possibility of cross-resistance between pyrethroids and DDT, the remaining adults from families that gave an indication of resistance to DDT were pooled and labelled as ROF (DDT-resistant Obuasi funestus). These were exposed to 0.75% permethrin for 1 h. Similarly, the remaining adults from those families that showed full susceptibility to DDT (showing 100% mortality) were pooled and labelled as SOF (DDT-susceptible Obuasi funestus). These were also exposed to 0.75% permethrin for 1 h. Samples of the survivor cohort (survivors to DDT or permethrin exposure) as well as a sample of susceptible mosquitoes from the DDT exposure assays were stored dry on silica gel and subsequently used for sodium channel gene analysis.

2.4. Biochemical analysis

Assays based on the method described by Penilla et al. (1998) were conducted to determine levels/activities of known detoxifying enzyme systems including monooxygenases, non-specific esterases and GSTs as well as to detect the presence of an altered acetylcholinesterase associated with carbamate resistance. Analysis included frozen female samples from 39 families. Sample sizes ranged from 3 to 19 females per family. The assays were arranged on 96-well microtitre plates where each microtitre plate contained samples from at least three families as well as ten *A. funestus* FANG (susceptible) females that were used for direct comparative purposes (two-sample *t*-tests) following adjustment for total protein content. Blanks were included on each plate as a negative control.

Table 1 Sequences of primers used in the study

Primer	Sequence (5′–3′)	T_M (°C)
N1	AAC GAT GGG TGC GTT AGG TA	60.4
N3	GTG CTA TGC GGA GAA TGG AT	60.4
N4	GTT GGT GCA GAC AAG GAT GA	60.4

T_M: melting temperature.

2.5. Molecular analysis of the IIS5–IIS6 domain of the sodium channel

2.5.1. Primer design

Three sets of primers (N1, N3 and N4) were used to amplify the IIS5–IIS6 domain of the sodium channel gene. Primers N1 (forward) and N4 (reverse) flank the region of interest, whilst N3 is an internal forward primer. Primer sequences are given in Table 1. PrimerSelect (DNASTAR Inc., Madison, WI, USA) and Primer 3 software (http://frodo.wi.mit.edu/cgi-bin/primer3/primer3_www.cgi) were used for primer design. PCR conditions were optimised for each primer pair (see Section 2.5.3 for details).

2.5.2. Genomic DNA

Genomic DNA was extracted according to the method of Collins et al. (1987) from single mosquitoes drawn from the insecticide exposure survivor and susceptible cohorts described in Section 2.3. DNA pellets were re-suspended in 100 μl of distilled water and stored at −20 °C.

2.5.3. PCR amplification

The IIS5–IIS6 segment of domain II of the sodium channel gene of each mosquito was amplified by PCR using primer sets N1–N4 and N3–N4. PCR was carried out using ~50 ng of DNA, 10× PCR buffer (100 mM Tris-HCl (pH 8.3), 500 mM KCl, 3.3 pmol of each primer, 25 mM dNTPs, 2 mM $MgCl_2$ and 0.5 U Takara *Taq* polymerase (code no. R001AM; Takara Bio Inc., Otsu, Shiga, Japan)). Thermocycling conditions consisted of an initial denaturation step of 94 °C for 2 min, 35 cycles of 94 °C for 30 s, 55 °C for 30 s and 72 °C for 30 s, with a final extension step at 72 °C for 5 min. The N1 and N4 primers gave a product of ~1442 bp and the N3 and N4 pair gave a band size of ~324 bp. PCR products were purified using a QlAquick Gel Extraction Kit (cat no. 28704; Qiagen, Crawley, UK) and were then used as a template for direct sequencing.

2.5.4. Sequencing and sequence analysis

PCR products were sent to Inqaba Biotechnical Industries (Pty) Ltd. (Pretoria, South Africa) for sequencing. Sequencing was carried out in both directions using the same primers used for PCR (N1, N3 and N4). Twelve DDT survivors, 11 permethrin survivors and 3 DDT-susceptible mosquitoes were analysed. Sequences were aligned using MegAlign software and were analysed using DNASTAR Lasergene software (DNASTAR Inc.).

The sequences obtained from the Ghanaian *A. funestus* samples were compared with those of the *A. funestus* pyrethroid-resistant (FUMOZ-R) (GenBank accession no. **DQ399298**) and susceptible (FANG) (GenBank accession no. **DQ399296**) laboratory strains as well as with that of *A. gambiae* (Weill et al., 2000). The *A. funestus* sodium channel sequence (GenBank accession no. **DQ534436**) was not used in the analysis because it is a consensus *A. funestus* sequence between samples drawn from southern African laboratory strains and West African populations.

3. Results

3.1. DDT susceptibility tests

Samples from 46 families were exposed to 4% DDT for 1 h and 28 showed 100% mortality 24 h post exposure. Ten families were scored as DDT resistant based on the WHO criterion that a final mortality 24 h post exposure of ≤80% indicates resistance (WHO, 1998). A further seven families gave final mortalities ranging between 80% and 99%. The mean percentage mortality 24 h post exposure across all families was 91.7%. The results are summarised in Table 2.

3.2. Permethrin susceptibility tests

Samples from the larger families as well as pooled samples of families scored as DDT resistant (ROF) or DDT susceptible (SOF) were exposed to 0.75% permethrin for 1 h (Table 2). All samples gave an indication of permethrin resistance, although only the ROF sample was scored as resistant according to WHO criteria.

3.3. Biochemical analysis

3.3.1. α esterase activity

Figure 1 shows mean levels of α esterase activity per family, with the reference FANG sample for each family shown adjacently. There was a significantly increased level of α esterase activity in 22 (56.4%) of 39 families. There was no significant correlation between α esterase activity and insecticide resistance phenotype by family ($P > 0.05$).

3.3.2. Monooxygenase activity

Figure 2 shows mean levels of monooxygenase activity per family, with the reference FANG sample for each family shown adjacently. There was a significantly increased level of monooxygenase in only 8 (20.5%) of 39 families. There was no significant correlation between monooxygenase level and insecticide resistance phenotype by family ($P > 0.05$).

3.3.3. Glutathione S-transferase activity

Figure 3 shows mean levels of GST activity per family, with the reference FANG sample for each family shown adjacently. There was a significantly increased level of GST activity in 10 (25.6%) of 39 families. There was no significant correlation between level of GST activity and insecticide resistance phenotype by family ($P > 0.05$).

3.3.4. Acetylcholinesterase inhibition

Table 3 gives the mean percentage propoxur inhibition of acetylcholinesterase by family. Thirty (76.9%) of 39 families showed a significantly lower percentage of inhibition of acetylcholinesterase by propoxur compared with the reference strain (FANG).

Table 2 WHO susceptibility tests using 4% DDT and 0.75% permethrin exposures for 1 h against F1 progeny of families generated from field populations of *Anopheles funestus* from Ghana[a]

Family	DDT				Permethrin			
	60-min KD	Dead at 24 h (*n*)	Total (*N*)	% mortality at 24 h	60-min KD	Dead at 24 h (*n*)	Total (*N*)	% mortality at 24 h
OF1[b]	13	12	16	75.0	10	9	10	90.0
OF8[b]	16	17	23	73.9				
OF14	16	16	18	88.9				
OF16	11	13	14	92.9				
OF17	21	22	23	95.7				
OF20[b]	12	9	13	69.2	10	9	10	90.0
OF24	23	23	24	95.8				
OF25	33	34	36	94.4	30	29	30	96.7
OF27[b]	21	20	29	69.0				
OF28[b]	12	11	14	78.6				
OF29	9	9	10	90.0				
OF32	17	19	20	95.0				
OF36[b]	19	22	29	75.9	16	15	16	93.8
OF37[b]	9	18	25	72.0				
OF39[b]	12	13	23	56.5				
OF42[b]	13	14	19	73.7				
OF43[b]	5	4	7	57.1				
ROF[c]					24	18	24	75.0
SOF[d]					25	23	25	92.0

[a] Sixty-minute knockdown (KD) and 24 h post exposure mortalities are given.
[b] Families showing <80% mortality 24 h post exposure to DDT.
[c] Pooled sample drawn from families showing significant resistance to DDT.
[d] Pooled sample drawn from families showing full susceptibility (100% mortality) to DDT.

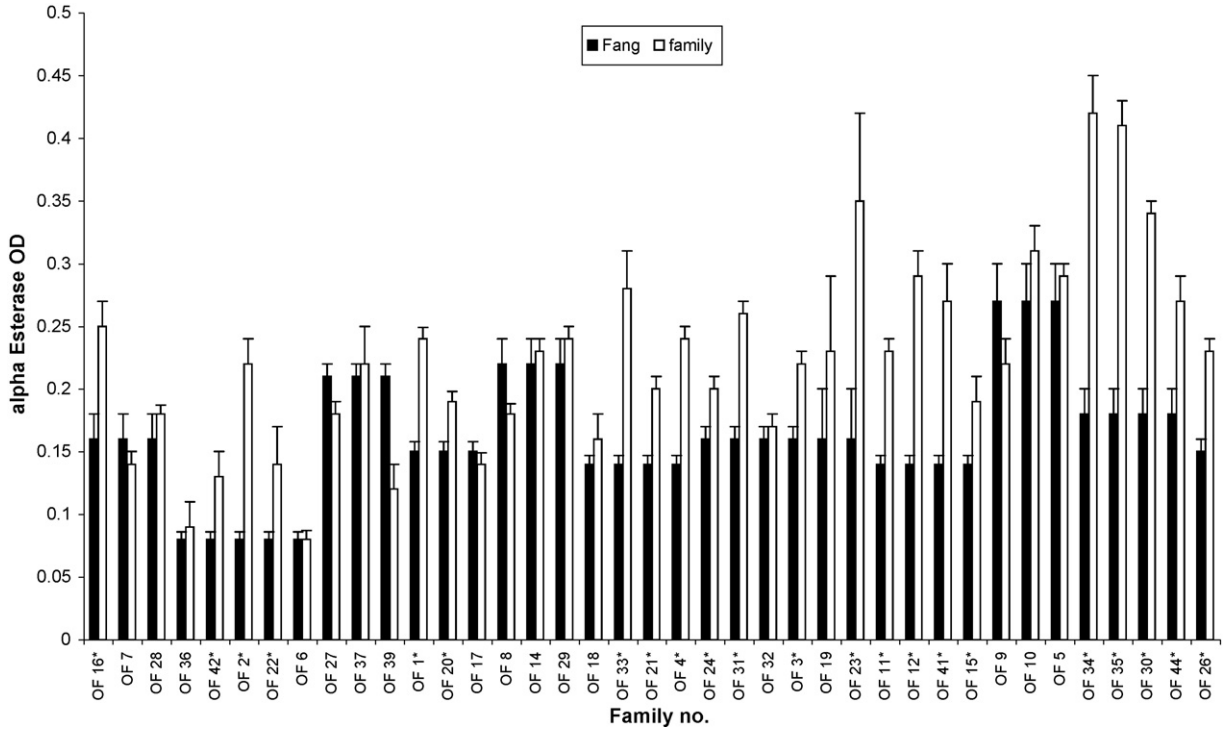


Figure 1 Levels of non-specific esterase by Ghanaian *Anopheles funestus* family using α-naphthyl acetate as a substrate. Esterase levels of *A. funestus* (FANG) reference samples are given adjacent to each family. * Indicates families with significantly increased levels of esterase activity following direct comparisons (two-sample *t*-tests) with associated FANG samples.

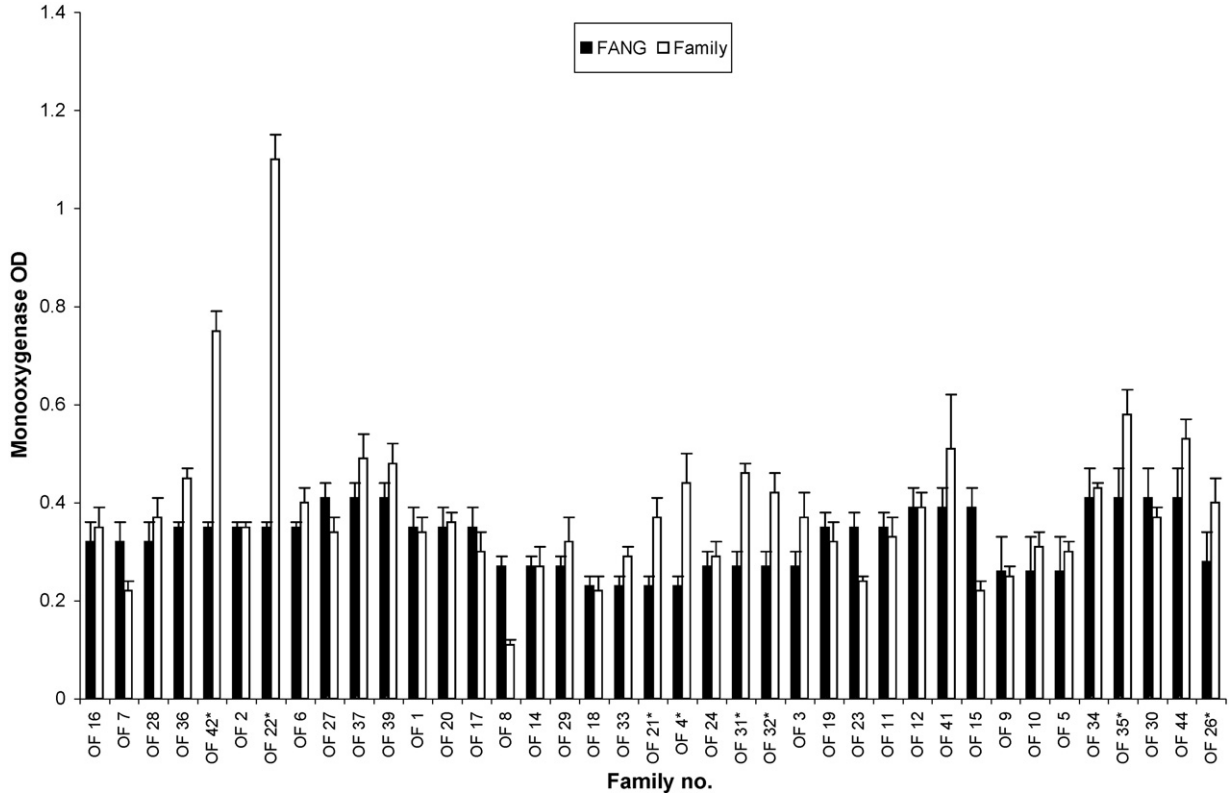


Figure 2 Levels of monooxygenase by Ghanaian *Anopheles funestus* family. Monooxygenase levels of *A. funestus* (FANG) reference samples are given adjacent to each family. * Indicates families with significantly increased levels of monooxygenase activity following direct comparisons (two-sample *t*-tests) with associated FANG samples.

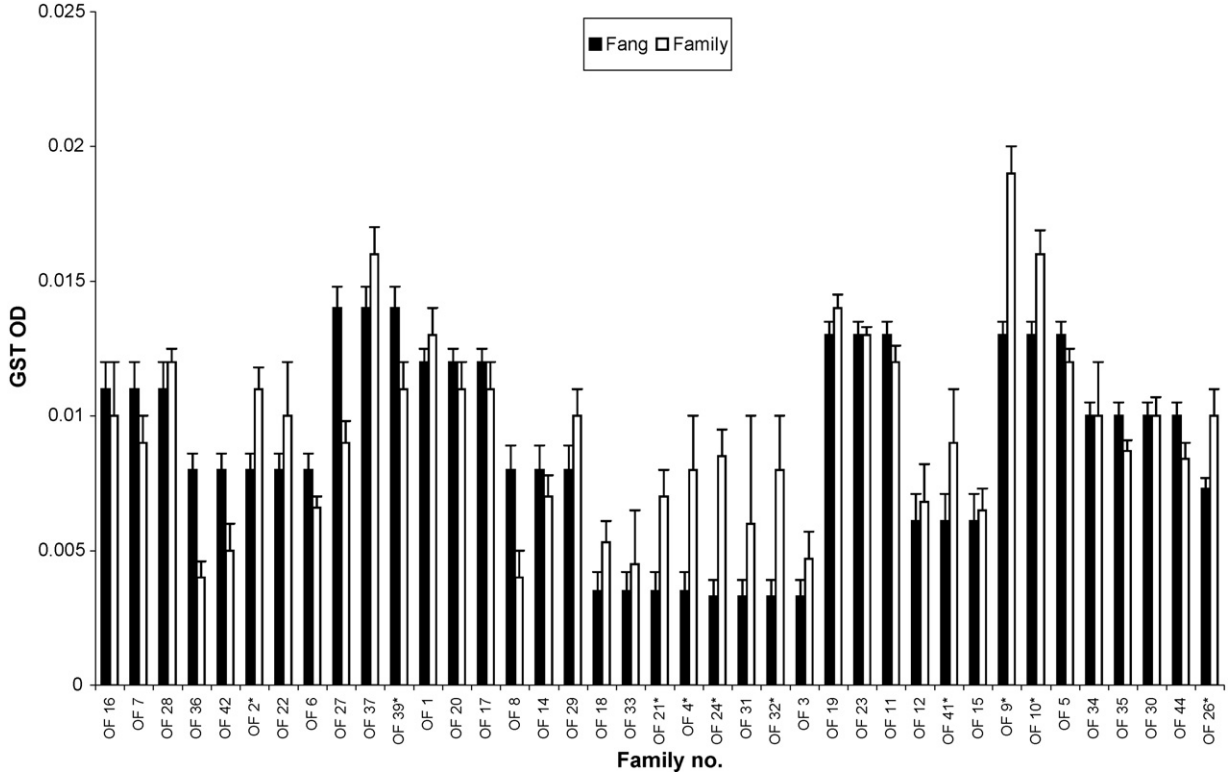


Figure 3 Levels of glutathione S-transferase (GST) activity by Ghanaian *Anopheles funestus* family. GST levels of *A. funestus* (FANG) reference samples are given adjacent to each family. * Indicates families with significantly increased levels of GST activity following direct comparisons (two-sample *t*-tests) with associated FANG samples.

Table 3 Mean percentage of acetylcholinesterase inhibition by propoxur of the F1 progeny of families of *Anopheles funestus* from Ghana

Family	Sample size (*n*)	Mean % inhibition ± SE
OF1	10	46.2 ± 3*
OF2	3	5.6 ± 5.6*
OF3	9	27.4 ± 2.7*
OF4	4	16.9 ± 4.8*
OF5	13	45.2 ± 3.9*
OF6	11	60.8 ± 5.7
OF7	12	48.1 ± 8.1
OF8	10	42 ± 2.5*
OF9	10	25 ± 2.7*
OF10	14	41 ± 3.3*
OF11	11	36 ± 4.6*
OF12	12	29 ± 4.1*
OF14	14	54.1 ± 3.5*
OF15	11	27.5 ± 1.7*
OF16	6	57.6 ± 3.7
OF17	11	54.5 ± 3.9*
OF18	11	31.2 ± 3.2*
OF19	15	32.3 ± 6.9*
OF20	6	14.9 ± 6.2*
OF21	11	26.7 ± 5.8*
OF22	6	61.1 ± 6.1
OF23	13	37.8 ± 4.6*
OF24	14	24.2 ± 6.5*
OF26	12	46.4 ± 4.2*
OF27	19	65 ± 3.3
OF28	7	51.4 ± 10.3
OF29	12	31.7 ± 3.3*
OF30	14	42.6 ± 3.9*
OF31	3	25 ± 3.6*
OF32	11	40.3 ± 4.2*
OF33	10	30.5 ± 1.6*
OF34	3	34.1 ± 1.6*
OF35	7	32.6 ± 4*
OF36	9	42 ± 4.9*
OF37	4	63.9 ± 2.7
OF39	12	57.3 ± 3.2
OF41	4	36.5 ± 6.7*
OF42	8	57.4 ± 6.1
OF44	9	27.8 ± 6.2*
FANG	10	66.1 ± 3

* Significantly lower inhibition than susceptible strain ($P < 0.05$).

3.4. Molecular analysis of the IIS5—IIS6 domain

PCR amplification of the IIS5—IIS6 domain of the sodium channel gene of DNA taken from F1 Ghanaian *A. funestus* yielded a 238 bp product using the N3 and N4 primers. The N1 and N4 primers yielded a product size of 1423 bp using the same DNA templates. The entire sequence has been deposited in GenBank (accession no. **EF687845**). Alignment of the sequences obtained from the DDT survivors, permethrin survivors and DDT-susceptible samples showed 100% homology (data not shown).

Supplementary Figure 1 shows the alignment of the genomic sequences of DDT/pyrethroid-resistant *A. funestus* from Ghana against the pyrethroid-resistant (FUMOZ-R) (GenBank accession no. **DQ399298**) and pyrethroid-susceptible (FANG) (GenBank accession no. **DQ399296**) laboratory strains of *A. funestus* as well as *A. gambiae* (obtained from Weill et al. (2000), no accession no. given). Alignment of the nucleotide sequences from the Ghanaian samples revealed a sequence similarity of 99.4%, 99.5% and 68.5% to FUMOZ-R, FANG and *A. gambiae*, respectively. Interestingly, there was a deletion of 5′ATAATGT3′ from position 99—105 (nucleotide position numbering of FUMOZ-R) in the Ghanaian *A. funestus* samples compared with the same position in FUMOZ-R and FANG. This same segment was inserted at position 1103—1109 (nucleotide position numbering of FUMOZ-R) in the Ghanaian samples, from where it was absent in FUMOZ-R. The *kdr* mutation (position 1291 indicated in Supplementary Figure 1), which has been reported in other insect species including *A. gambiae* (Martinez-Torres et al., 1998; Ranson et al., 2000), *A. arabiensis* (Matambo et al., 2007; Verhaeghen et al., 2006) and *A. sacharovi* (Lüleyap et al., 2002), was absent in the Ghanaian *A. funestus* population as well as in FUMOZ-R and FANG.

4. Discussion

Based on the WHO (1998) insecticide susceptibility protocol and data analysis guidelines, several F1 families derived from wild-caught *A. funestus* females from Ghana showed evidence of resistance to DDT. Further analysis of pooled and remaining familial samples against permethrin showed evidence of pyrethroid resistance in the parent population and hints at the possibility of cross-resistance between pyrethroids and DDT. The data presented by Coetzee et al. (2006), based on assays against wild-caught mosquitoes, indicate a significant level of resistance to DDT (60.9% final mortality) but suggest full susceptibility to pyrethroids (100% final mortality recorded against the pyrethroids deltamethrin and cyfluthrin). This discrepancy may reflect differences in the age at which samples were exposed to insecticide. All F1 samples assayed in this study were 2—3 days old, whilst the data presented in Coetzee et al. (2006) reflect the responses of wild-caught mosquitoes of unknown, presumably variable, age and therefore variable physiological status.

Cross-resistance between DDT and pyrethroids is well documented in *A. gambiae* (see reviews by Chandre et al., 1999; Hemingway and Ranson, 2000) and is closely associated with *kdr* mutations in this species (Martinez-Torres et al., 1998; Ranson et al., 2000). Recently Xu et al. (2006) found a strong correlation between *kdr* allelic expression and insecticide resistance. Data presented here show that those mutations previously defined as *kdr* are not present in the Ghanaian *A. funestus* population studied. Furthermore, no alternative mutations were identified in the sodium channel gene region analysed that could be correlated with the resistance phenotypes recorded. It is possible that previously undescribed *kdr*-type mutations may be present in regions of the sodium channel locus that were not sequenced.

Based on these considerations, it is most likely that insecticide resistance in this population is mediated by metabolic sequestration or detoxification. This is because

biochemical profiling of candidate detoxification enzyme systems of F1 progeny by family does show evidence of esterase, monooxygenase and GST elevation. However, the absence in these data of any clear correlations between enzyme levels and resistance phenotypes across families precludes any definitive identification of the controlling mechanism(s) of resistance. The absence of correlations between bioassay and biochemical data may present as the effect of enzyme upregulation for physiological reasons that are unrelated to the expression of insecticide resistance. Alternatively, real correlations in this data set may be masked by the broad range of substrate specificity inherent in the design of the biochemical assays used here. For example, the titration of monooxygenase activity covers the full spectrum of cytochrome P450s active in the crude homogenate, of which only a very small proportion are likely to be associated with insecticide detoxification.

Most families gave indications of reduced acetylcholinesterase sensitivity to propoxur inhibition. Unfortunately, additional insecticide exposures to carbamates and organophosphates were not possible owing to the limited number of F1 progeny obtained from each family. However, Coetzee et al. (2006), using the standard WHO insecticide susceptibility method against wild-caught mosquitoes, showed that the Obuasi *A. funestus* population also carries resistance to the carbamate bendiocarb. They recorded a final mortality of 71.4% 24 h post exposure to 0.1% bendiocarb. Altered acetylcholinesterases are generally considered to confer high levels of resistance to carbamates even in those individuals that are heterozygous at this locus (Hemingway and Ranson, 2000). The production of this resistance phenotype may be further enhanced by increased esterase activity.

The results presented here can be used to plan an effective malaria control strategy in the Obuasi region. The resistance phenotypes characterised for *A. funestus* need to be considered in conjunction with those identified in *A. gambiae* from the same region. Coetzee et al. (2006) demonstrate DDT, pyrethroid and carbamate resistance in *A. gambiae* S form samples. An integrated vector management system is proposed. This would include rotational use of insecticides for indoor residual spraying, the use of insecticide-treated bed nets, house screening, larval control and environmental management.

Authors' contributions: MC and RHH conceived the study; PNO conducted the molecular and biochemical experiments and analysed the data under the guidance of BDB and LLK; PNO and BDB wrote the manuscript and MC, LLK and RHH critically reviewed it; RHH provided the Ghanaian *A. funestus* samples; LLK and BDB provided bioassay data. All authors read and approved the final manuscript. BDB is guarantor of the paper.

Acknowledgements: Dr P. van Wyk, AngloGold/Ashanti Health Services, is thanked for facilitating the visit to Obuasi and the collection of mosquitoes. The reviewers of the first draft of this manuscript are thanked for their advice.

Funding: Medical Research Council of South Africa and the National Research Fund.

Conflicts of interest: None declared.

Ethical approval: Not required.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.trstmh.2008.02.022.

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